

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn import datasets
from sklearn.preprocessing import StandardScaler, LabelEncoder
```

1

```
from sklearn.datasets import load_iris
```

2

```
# Load dataset from sklearn
```

```
iris = datasets.load_iris()
```

```
# Create DataFrame
```

```
df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
```

```
# Add target column
```

```
df['species'] = iris.target
```

```
# View first 5 rows
```

```
print(df.head())
```

3

```
print(df.isnull().sum())
```

4

```
print(df.info())
```

```
print(df.describe())
```

5

```
df.drop_duplicates(inplace=True)
```

6

```
# Map numeric target to species names
```

```
df['species'] = df['species'].map({
```

```
    0: 'setosa',
```

```
    1: 'versicolor',
```

```
    2: 'virginica'
```

```
})
```

```
print(df.head())
```

7

```
scaler = StandardScaler()
```

```
features = df.drop('species', axis=1)
```

```
scaled_features = scaler.fit_transform(features)
```

```
df_scaled = pd.DataFrame(scaled_features, columns=features.columns)
```

```
# Add species back
```

```
df_scaled['species'] = df['species']
```

```
print(df_scaled.head())
```

8

```
encoder = LabelEncoder()
```

```
df_scaled['species_encoded'] = encoder.fit_transform(df_scaled['species'])
```

```
print(df_scaled.head())
```

9

```
print(df_scaled.head())
```